

Axioms of Probability and Probability Limit Concepts

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I Axioms of Probability

A set function P on the σ -field $\mathcal{F}$ is a *probability* or *probability measure* if it satisfies the following conditions:

- (a) Non-Negativity: $P(E) \geq 0$ for all $E \in \mathcal{F}$.
- (b) Unit Measure: $P(\Omega) = 1$.
- (c) Countable Additivity: If $E_1, E_2, \dots \in \mathcal{F}$ are disjoint, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n).$$

II Properties of Probability

a. Subtraction Law

The Subtraction Law of Probability states that

$$P(F - E) = P(F) - P(E)$$

Proof.

Let $F, E \in \mathcal{F}$ be events such that $E \subset F$. Then, we have that $F = E \cup (F - E)$. Then, since P is a real valued function, $P(F) = P(E) \cup P(F - E)$. Then, by the Axiom of countable additivity, $P(F) = P(E) + P(F - E)$ so $P(F - E) = P(F) - P(E)$. $\square$

Remark. Note that by the Axiom of Non-Negativity, $P(F - E) = P(F) - P(E) \geq 0 \implies P(F) - P(E)$. Further if $F = E$ then $P(\emptyset) = 0$. Finally, by the Axiom of Unit Measure, $P(E^c) = 1 - P(E)$.

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b. Addition Law

The Addition Law of Probability states that

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Proof. Let $F, E \in \mathcal{F}$ be events such that $E \subset F$. Then, $E \cup F = (E - E \cap F) \cup (E \cap F) \cup (F - E \cap F)$. Since P is a real valued function, $P(E \cup F) = P(E - E \cap F) \cup P(E \cap F) \cup P(F - E \cap F)$. By the Law of Subtraction and the Axiom of Countable Additivity, $P(E \cup F) = P(E) - P(E \cap F) + P(E \cap F) + P(F - E \cap F)$. Canceling terms, $P(E \cup F) = P(E) + P(F) - P(E \cap F)$. $\square$

III Limit Concepts in Probability

Let $\{E_n\}$ be a sequence of events. We say that $\{E_n\}$ is *monotone* when

$$E_1 \subset E_2 \subset \cdots \quad \text{or} \quad E_1 \supset E_2 \supset \cdots$$

For a monotone sequence $\{E_n\}$ of events, we define

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{n=1}^{\infty} E_n \quad \text{or} \quad \lim_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} E_n,$$

depending upon whether the sequence is increasing or decreasing.

Example Let $\{x_n\}$ be a sequence of numbers. If $x_n \uparrow x_\infty$, then the sequence $\{(-\infty, x_n]\}$ of subsets of $\mathbb{R}$ is monotonically increasing, and we have

$$\lim_{n \rightarrow \infty} (-\infty, x_n] = (-\infty, x_\infty).$$

For any $x < x_\infty$ we have $x \in (-\infty, x_n]$ for all large enough n , and since $x_n \uparrow x_\infty$, therefore $x \in \bigcup_n (-\infty, x_n]$. However, $x_\infty \notin (-\infty, x_n]$ for any n , and so $x_\infty \notin \bigcup_n (-\infty, x_n]$.

If $x_n \downarrow x_\infty$, then $\{(-\infty, x_n]\}$ is monotonically decreasing, and it follows that

$$\lim_{n \rightarrow \infty} (-\infty, x_n] = (-\infty, x_\infty],$$

since for any $x \leq x_\infty$ we have $x \in (-\infty, x_n]$ for all n , and therefore $x \in \bigcap_n (-\infty, x_n]$.

The limit of any monotone sequence of events is itself an event. The following theorem shows how we may obtain the probability of a limit event.

Example Let $E_n = (-\infty, \frac{-1}{n}]$. Then,

$$\lim_{n \rightarrow \infty} \left(-\infty, \frac{-1}{n}\right] = (-\infty, 0],$$

Because $0 \notin E_n$. If we consider the same problem but with $\frac{1}{n}$, then E_n is monotone decreasing and $0 \in E_n$ so the limit is $(-\infty, 0]$.